

ISYE 671: Linear Optimization and Network Flows — Supplemental Note

Revised Exercise 2.4: Swedish Steel — Standard vs. Premium Product Selection

Instructor: Dr. Ziteng Wang | **Date:** February 9, 2026

Motivation: Why the Original Exercise Needs Revision

In the Swedish Steel fixed-charge model (Section 2.3 of the lecture notes), the standard product requires nickel content between 30 and 35 kg (3.0%–3.5% of the 1000 kg charge). If we naively add a “premium” option requiring at least 50 kg of nickel ($\geq 5\%$), we face a contradiction: the standard upper bound ($N \leq 35$) and the premium lower bound ($N \geq 50$) cannot hold simultaneously. A simple bonus variable cannot resolve this — we need to model a genuine **either-or product selection** using disjunctive constraints.

This revised exercise illustrates a powerful and broadly applicable IP technique: using Big-M constraints to activate one of several mutually exclusive specification sets.

Revised Problem Statement

Swedish Steel can produce one of two product grades from the same furnace charge:

- **Standard grade:** Nickel content between 30 and 35 kg (as in the original model). Sells at the base price.
- **Premium grade:** Nickel content at least 50 kg. Earns a bonus of 2,000 kroner but incurs a testing/certification fee of 500 kroner.

All other chemical specifications (carbon, chromium, molybdenum) remain identical for both grades. The fixed-charge setup structure for ingredients 1–4 also remains unchanged.

Task: Modify the fixed-charge Swedish Steel model to allow the company to choose which grade to produce, and determine the optimal decision.

Solution

Step 1: Define New Decision Variables

We introduce two new binary variables, consistent with the notation in the Exercise Solutions document:

$$z = \begin{cases} 1 & \text{if the company opts into the premium contract} \\ 0 & \text{otherwise} \end{cases}$$

$$w = \begin{cases} 1 & \text{if the premium nickel specification is active (i.e., premium is both chosen and met)} \\ 0 & \text{otherwise} \end{cases}$$

The variable z represents the **business decision** to pursue the premium contract. The variable w represents the **technical outcome** that the premium specification is actually enforced. We require $w \leq z$: the premium specification can only be active if the company has opted in.

Why two variables? The testing fee of 500 kroner is paid whenever the company opts in ($z = 1$), regardless of whether the nickel target is achieved. The bonus of 2,000 kroner is earned only when the premium specification is met ($w = 1$). Separating the opt-in decision from the specification enforcement allows the objective to capture these distinct cost/benefit components.

Step 2: Replace the Nickel Constraints with Disjunctive Constraints

Let N denote the nickel content of the charge:

$$N = 0.180x_1 + 0.032x_2 + x_5$$

The original model has two nickel constraints: $N \geq 30$ and $N \leq 35$. We must replace these with constraints that enforce *either* the standard specification $[30, 35]$ (when $w = 0$) or the premium specification $N \geq 50$ with no upper bound (when $w = 1$).

Let M be a sufficiently large constant. Since N cannot exceed the total charge weight, $M = 1000$ is valid (though tighter values are preferable in practice — see Discussion Point 1 below).

Standard nickel lower bound (active when $w = 0$, relaxed when $w = 1$):

$$N \geq 30 - Mw$$

Standard nickel upper bound (active when $w = 0$, relaxed when $w = 1$):

$$N \leq 35 + Mw$$

Premium nickel lower bound (active when $w = 1$, relaxed when $w = 0$):

$$N \geq 50 - M(1 - w)$$

Note that the premium grade has **no upper bound** on nickel content, so there is no fourth constraint. This is an asymmetric disjunction: one side has two bounds, the other has one.

Verification that these work correctly:

w	Std lower	Std upper	Prem lower
0 (standard)	$N \geq 30$ (binds)	$N \leq 35$ (binds)	$N \geq 50 - M$ (trivial)
1 (premium)	$N \geq 30 - M$ (trivial)	$N \leq 35 + M$ (trivial)	$N \geq 50$ (binds)

These four constraints **replace** the original two nickel constraints. All other constraints in the model (weight, carbon, chromium, molybdenum, switching, setup) remain unchanged.

Step 3: Link w and z

The premium specification can only be active if the company has opted in:

$$w \leq z$$

This constraint ensures that if $z = 0$ (no opt-in), then $w = 0$ (standard specification is enforced).

Step 4: Modify the Objective Function

The objective captures two separate financial effects:

- **Testing fee:** 500 kroner, paid whenever $z = 1$ (opt-in), regardless of outcome.
- **Premium bonus:** 2,000 kroner, earned whenever $w = 1$ (premium spec met and opted in).

$$\min \quad \underbrace{16x_1 + 10x_2 + 8x_3 + 9x_4 + 48x_5 + 60x_6 + 53x_7}_{\text{variable ingredient costs}} + \underbrace{350(y_1 + y_2 + y_3 + y_4)}_{\text{setup costs}} + \underbrace{500z}_{\text{testing fee}} - \underbrace{2000w}_{\text{premium bonus}}$$

Key modeling insight: Since we are minimizing and w carries a negative coefficient (-2000), the optimizer *wants* to set $w = 1$ whenever possible. This means:

- We do not need constraints that *force* $w = 1$ when premium conditions are met — the optimizer does it automatically.
- We only need constraints that *prevent* $w = 1$ when conditions are not met: the four disjunctive nickel constraints and the linking constraint $w \leq z$.

Conversely, z carries a positive coefficient ($+500$), so the optimizer prefers $z = 0$. It will set $z = 1$ only when the benefit from $w = 1$ (bonus of 2,000) outweighs the testing fee (500) plus any additional ingredient cost.

Step 5: Complete Revised Model

$$\min \quad \sum_{j=1}^7 c_j x_j + \sum_{j=1}^4 350 y_j + 500 z - 2000 w$$

subject to:

$$\sum_{j=1}^7 x_j = 1000 \quad (\text{weight})$$

$$6.5 \leq 0.0080x_1 + 0.0070x_2 + 0.0085x_3 + 0.0040x_4 \leq 7.5 \quad (\text{carbon, unchanged})$$

$$N \geq 30 - Mw \quad (\text{standard nickel lower})$$

$$N \leq 35 + Mw \quad (\text{standard nickel upper})$$

$$N \geq 50 - M(1 - w) \quad (\text{premium nickel lower})$$

where $N = 0.180x_1 + 0.032x_2 + x_5$

$$10 \leq 0.120x_1 + 0.011x_2 + x_6 \leq 12 \quad (\text{chromium, unchanged})$$

$$11 \leq 0.001x_2 + x_7 \leq 13 \quad (\text{molybdenum, unchanged})$$

$$x_j \leq u_j y_j \quad \text{for } j = 1, \dots, 4 \quad (\text{switching/setup})$$

$$w \leq z \quad (\text{premium spec requires opt-in})$$

$$x_j \geq 0, \quad y_j \in \{0, 1\} \quad (j = 1, \dots, 4), \quad z \in \{0, 1\}, \quad w \in \{0, 1\}$$

Discussion Points for Class

1. Tightening Big-M Values

The value of M must be large enough to fully relax the inactive constraints. Here $M = 1000$ works because the nickel content is bounded by the total charge weight. However:

- **Too small** an M may cut off feasible solutions.
- **Too large** an M weakens the LP relaxation, making the problem harder for branch-and-bound.
- **Best practice:** Use the tightest valid M . For the standard lower bound $N \geq 30 - Mw$, we need $30 - M \leq 0$ when $w = 1$, so $M \geq 30$ suffices. For the standard upper bound $N \leq 35 + Mw$, we need the right-hand side to exceed the maximum feasible N when $w = 1$; since the premium spec pushes $N \geq 50$, a value of M such that $35 + M$ exceeds the maximum possible nickel content is needed — $M \geq 1000 - 35 = 965$ in the worst case, though tighter analysis using ingredient availabilities can reduce this. Students can derive the tightest valid M for each of the three constraints as a follow-up exercise.

2. The Relationship between z and w

In this model, the optimal solution will always have $z = w$ (either both 0 or both 1). Why? If $z = 1$ and $w = 0$, the company pays the testing fee but earns no bonus and still must meet the standard spec — strictly worse than $z = 0, w = 0$. The optimizer will never choose this.

So why keep both variables? Because the problem *statement* distinguishes the business decision (opt-in) from the technical outcome (spec met). In a more complex variant — for example, if the testing fee were paid upfront and the bonus depended on a stochastic yield — z and w could take different values. The two-variable formulation is the more general and extensible approach.

3. Generalization: Multiple Product Grades

If the company could choose among K grades (e.g., standard, premium, ultra-premium), we would introduce $w_k \in \{0, 1\}$ for $k = 1, \dots, K$ with:

$$\sum_{k=1}^K w_k = 1 \quad (\text{exactly one grade produced})$$

Each element's bounds would be controlled by the corresponding w_k via Big-M. This pattern arises frequently in process engineering and product design optimization.

4. Connection to Other Lecture Topics

This exercise ties together three techniques from the lecture:

- **Fixed-charge variables and switching constraints** (Section 2.3) — the setup costs for ingredients.
- **Mutual exclusivity** (Section 3.3, NASA) — exactly one product grade must be chosen.
- **Big-M disjunctive constraints** (Section 7.2, scheduling) — the either-or specification logic.

Students should recognize that the Big-M disjunction used here for product specifications is structurally identical to the Big-M disjunction used in scheduling for conflict constraints. The

underlying principle is the same: use a binary variable and a large constant to activate exactly one of two constraint sets.